CIS437 - Cloud Computing

– Cloud Load Balancing –

Distributing Traffic for Scale and Availability

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Based on the original version by Professor Erik Fredericks

Last Class

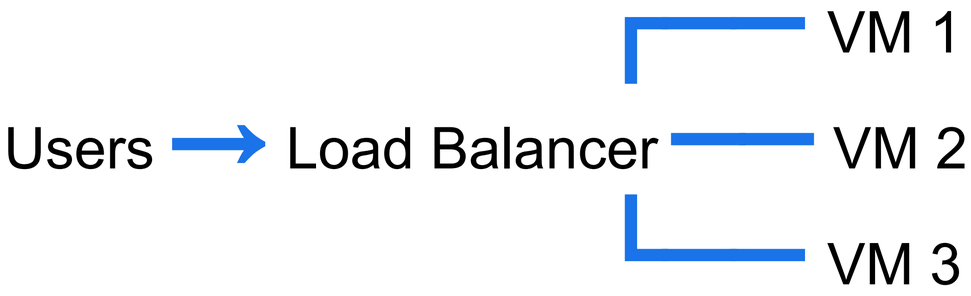
- **Shared VPC** lets multiple projects use a common centrally managed VPC.
- **VPC Network Peering** privately connects separate VPC networks.
- VPC Peering is **not transitive**.
- **Cloud VPN** provides encrypted connectivity over the Internet.
- **Dedicated Interconnect** provides a direct physical connection to Google.
- **Partner Interconnect** provides connectivity through a service provider.
- Choose the connection method based on network relationship, security, capacity, availability, and cost.

Agenda

- Why Load Balancing?
- Load Balancer Types: Application vs. Network
- Frontends, Backends, and Health Checks
- Load Balancing with MIGs and Autoscaling
- External vs. Internal Load Balancing
- Hands-on Lab
- Practice

Distribute traffic across multiple backend instances
Improve availability by avoiding unhealthy backends
Support scalability as application demand changes

Why Load Balance?





The load balancer distributes traffic; it does not itself create more VMs.

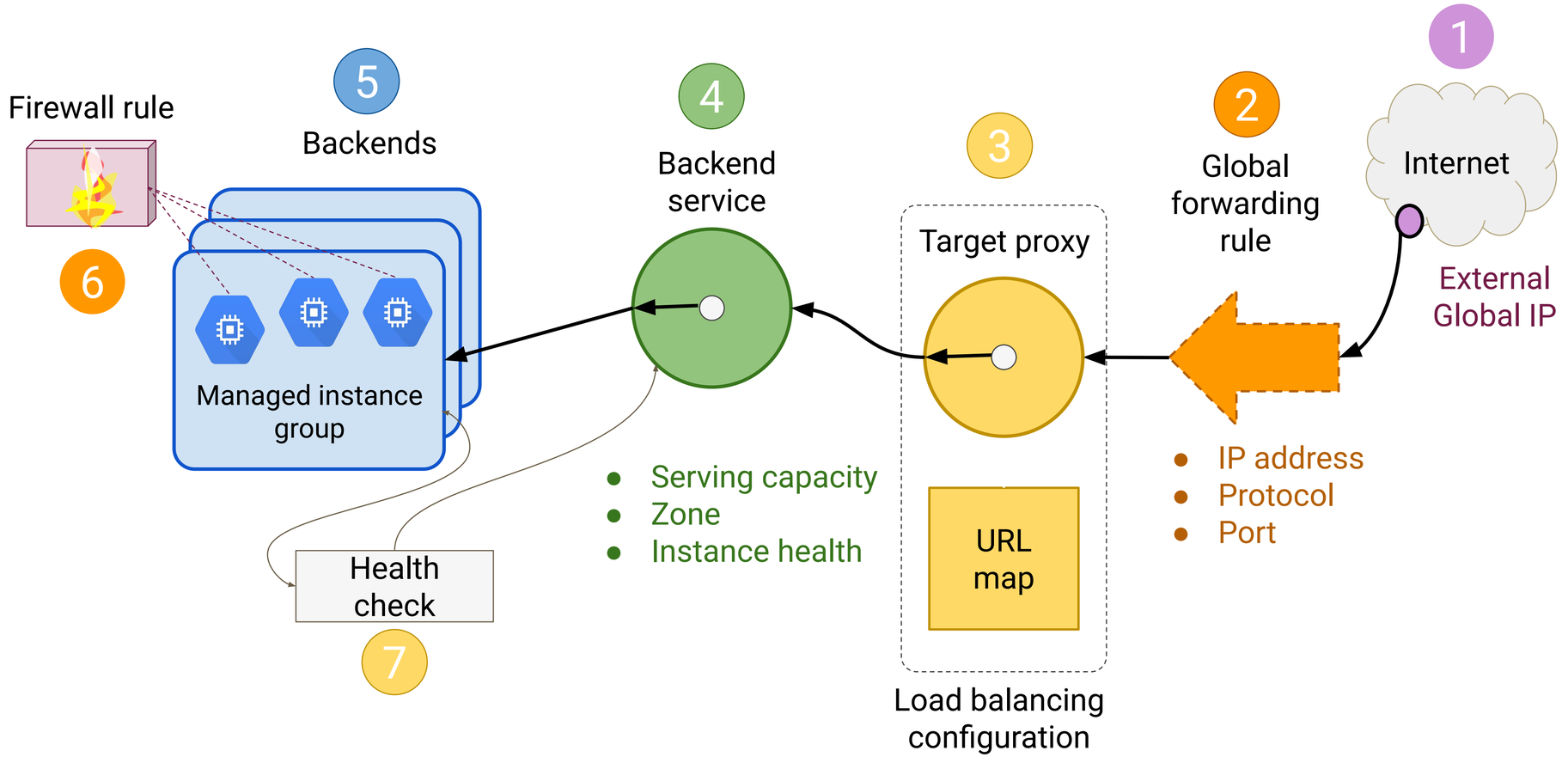
Types of Cloud Load Balancing

Type	Layer / traffic	Main idea
Application Load Balancer	Layer 7 — HTTP/HTTPS	Understands web requests; can route by URL/host
Proxy Network Load Balancer	Layer 4 — TCP/TLS	Proxy-based network traffic balancing
Passthrough Network Load Balancer	Layer 4 — TCP/UDP/other IP protocols	Preserves client packet information; backend terminates connection

Application LB = Layer 7 / HTTP(S)
Network LB = Layer 4 / TCP, UDP, or other network traffic

Official Doc: Cloud Load Balancing overview

How an HTTPS Request Flows



Client → IP / Forwarding Rule → Target Proxy → URL Map → Backend Service → Backend VM

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Frontend: Where Clients Connect

- Frontend is the client-facing entry point
- Includes an IP address, protocol, and port
- A forwarding rule receives matching traffic
- Example: external IP + HTTPS/443

Frontend = IP address + protocol + port

Backend: Where Requests Are Served

- Backends are the resources that handle requests
- For this class, think mainly of managed instance groups
- A backend service connects the load balancer to its backends
- Depending on LB type, backends may exist across zones or regions

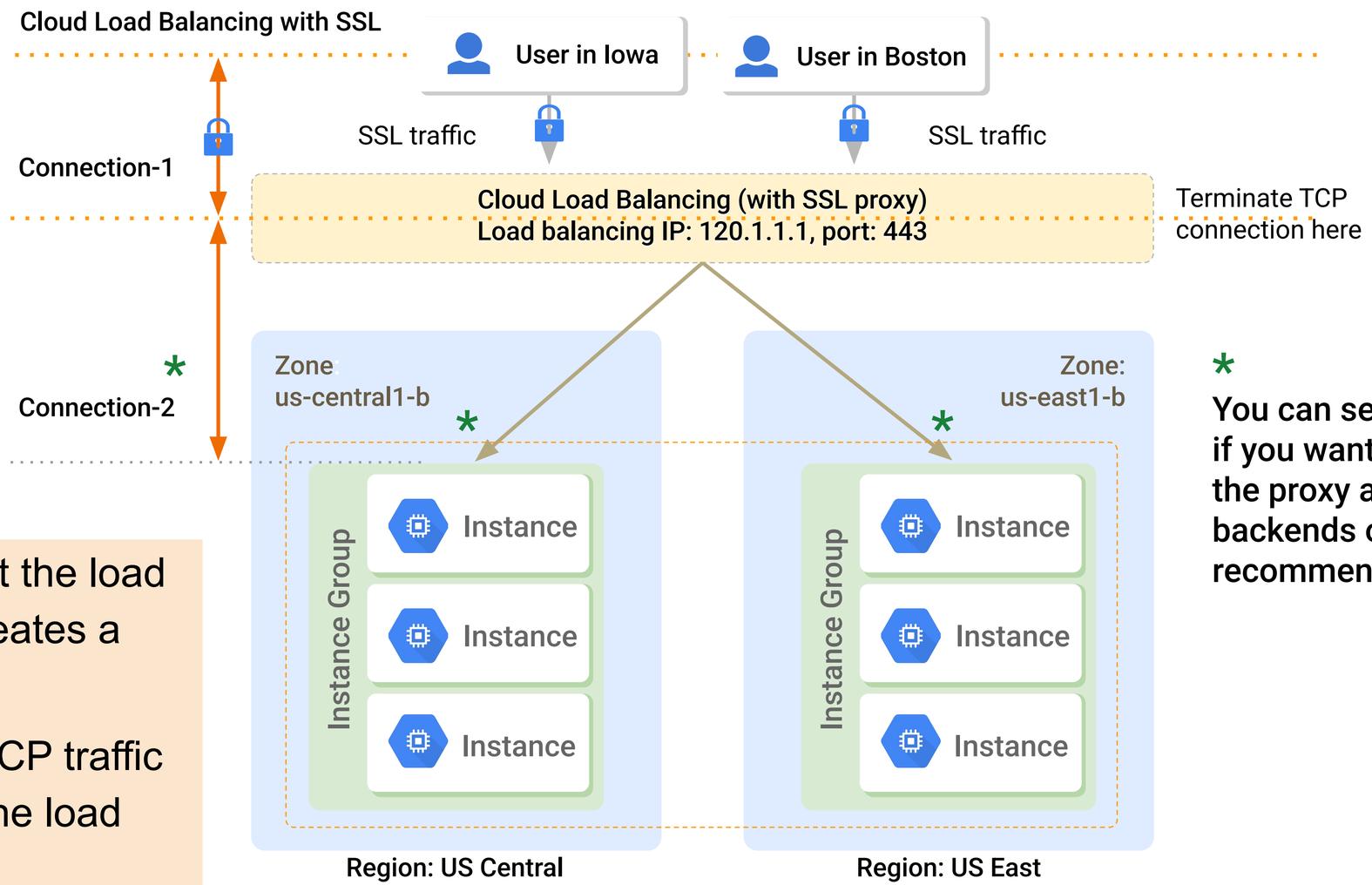
Backend = the resources that actually process the request

Health Checks: Can This Backend Serve Traffic?

- Periodically test whether backends are healthy
- New traffic is sent only to healthy, eligible backends
- Health-check traffic must be allowed through applicable firewall rules
- A health check detects problems; it does not itself add/remove VMs

Health check = “Is this backend ready to receive traffic?”

Proxy Network Load Balancer



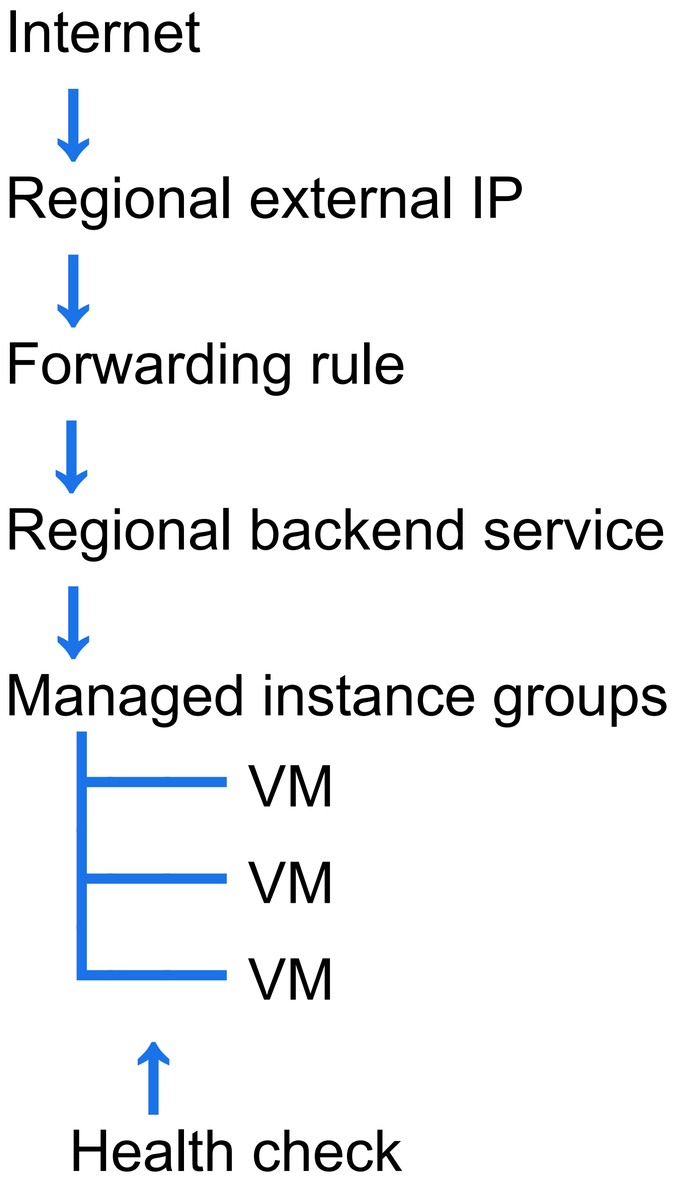
Client connection terminates at the load balancer; the load balancer creates a new connection to a backend.

- Layer 4 load balancing for TCP traffic
- Can terminate TLS/SSL at the load balancer

* You can separately decide if you want SSL between the proxy and your backends or not. Google recommends using SSL.

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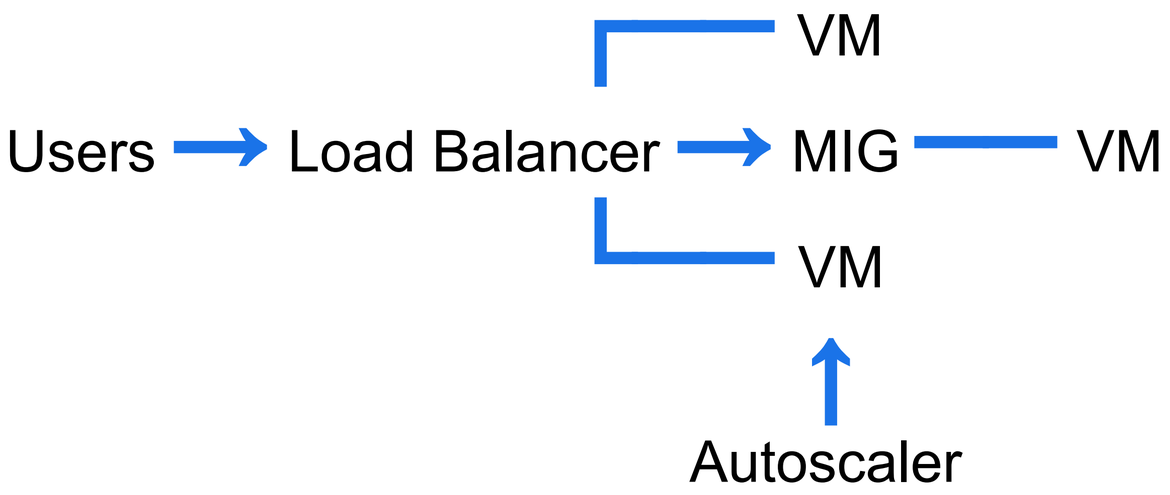
Regional Passthrough Network Load Balancer



- Layer 4, regional load balancing
- Preserves the original client source and destination IP information
- Backends can be managed instance groups in the same region

Load Balancing and Autoscaling

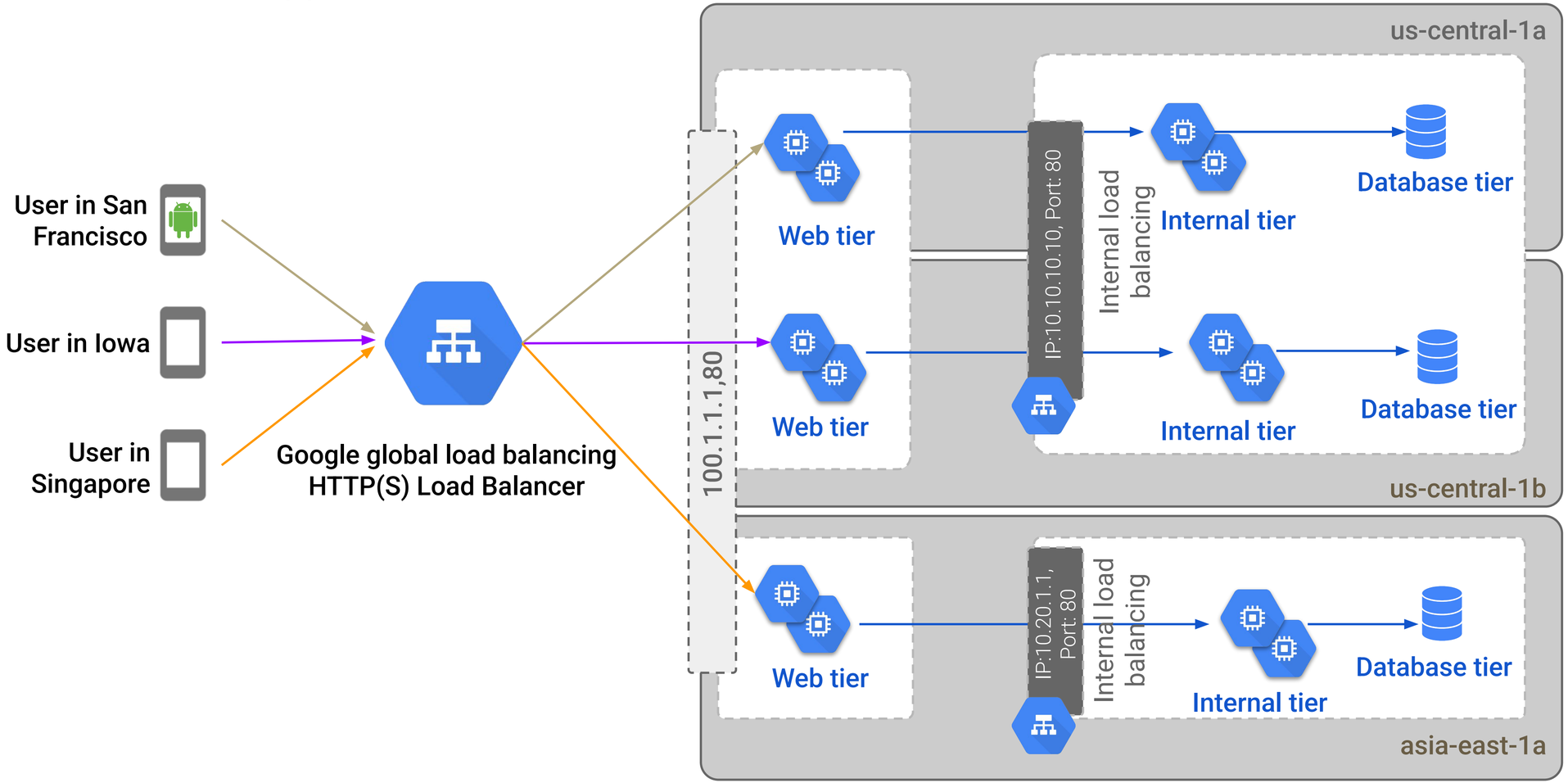
- **Load Balancer:** distributes requests across healthy backend VMs
- **Autoscaler:** increases or decreases the number of VMs based on demand
- **Health Check:** determines whether a backend is healthy
- **MIG:** manages a consistent group of VM instances



Load balancing handles traffic; autoscaling handles capacity.

External vs. Internal Load Balancing

External load balancing serves users coming from outside the VPC. Internal load balancing serves clients inside the private cloud network. A real application can use both



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Lab: HTTP Load Balancer with Cloud Armor

Configure an HTTP load balancer with global backends and stress test the load balancer and blocklist the stress test IP with Cloud Armor. The lab can be found here.



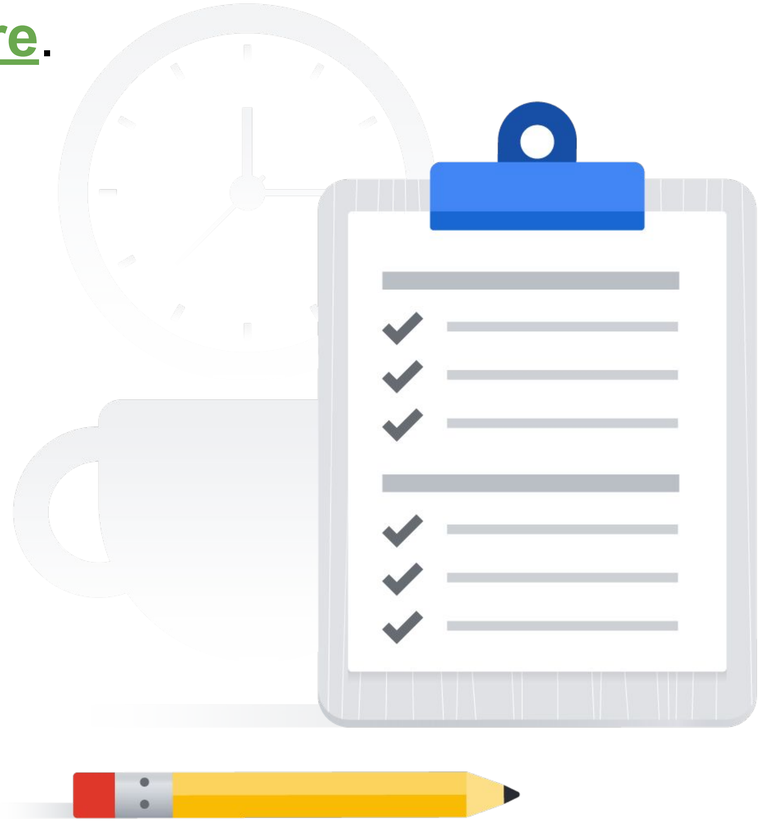
Lab objectives

- Create HTTP and health check firewall rules.
- Configure two instance templates.
- Create two managed instance groups.
- Configure an HTTP load balancer with IPv4 and IPv6.
- Stress test an HTTP load balancer.
- Blocklist an IP address to restrict access to an HTTP load balancer.

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Lab: Create an Internal Load Balancer

Set up an HTTP global load balancer and learn how load balancing can help scale your applications on Compute Engine. The lab can be found here.



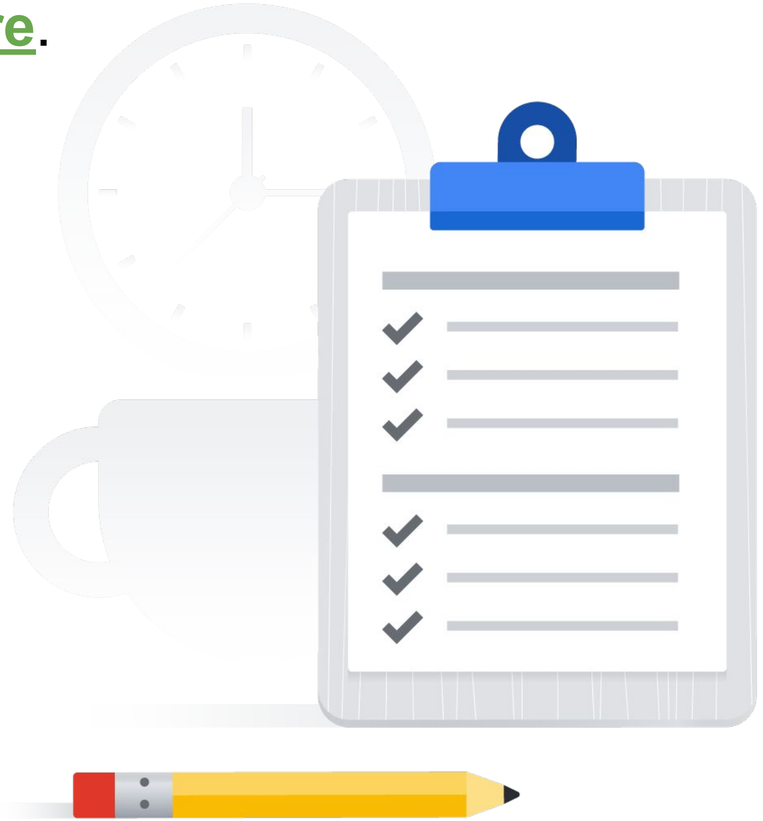
Lab objectives

- Create firewall rules for application and health-check traffic
- Configure two instance templates and managed instance groups
- Configure an internal load balancer
- Test private traffic through the load balancer

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Lab: Internal Load Balancer (Alternative)

Create a public-facing web server to serve the result of several "complex" calculations, in this case, calculating prime numbers. The lab can be found here.



Lab objectives

- Create a managed instance group of backends.
- Point an internal load balancer to the backends.
- Test the internal load balancer, and call it from a public facing web server.

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Summary

- Load balancing distributes traffic across multiple backends.
- Application Load Balancers handle HTTP/HTTPS traffic at Layer 7.
- Network Load Balancers handle Layer 4 network traffic.
- The frontend is where clients connect; backends serve the requests.
- Health checks help ensure new traffic goes to healthy backends.
- Load balancers distribute traffic; autoscalers adjust capacity.
- Load balancers can be external or internal depending on who needs access.

Practice: Design a Scalable Web Application

Scenario

A company runs a public web application on a managed instance group.
Traffic varies during the day, and one VM may occasionally fail.

1. Which type of load balancer is appropriate for public HTTP/HTTPS traffic?
2. What determines whether a VM should receive new requests?
3. What adds or removes VM instances as demand changes?
4. If the application should only be reachable from inside the VPC, which type of load balancer should we use?

